Ship Detection in SAR Images by Aggregating Densities of Fisher Vectors: Extension to a Global Perspective

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***Abstract*— Fisher vectors (FVs) can capture multiple order information from superpixels (SPs) in synthetic aperture radar (SAR) images. Existing FV-based ship detectors mainly exploit the local contrast of FVs (LCFVs) but do not consider their global density features. This may lead to degraded performance in terms of discrimination between ship targets and the complex sea clutter. In this article, two new global cues from FVs are designed based on the fact that target FVs exhibit much lower densities than those of clutter FVs and also have large distances to the latter. Our two new global cues can suppress the sea clutter and significantly enhance ship targets throughout the SAR image. We also design an improved local cue from FVs for ship detection, in which the intensity contrast of SPs is incorporated into the existing LCFV indicator to reduce false alarms. By fusing the above two new global cues (and an improved local cue from FVs), we propose a new method for ship detection in SAR images. Experimental results based on Gaofen-3 SAR images show that the newly proposed detector provides better detection performance than other state-of-the-art detectors, especially in the presence of strong and highly heterogeneous sea clutter.**

***Index Terms*— Density, Fisher vectors (FVs), ship detection, superpixels (SPs), synthetic aperture radar (SAR).**

I. Introduction

THE task of ship target detection is important in a number of marine surveillance applications, including maritime fishery, commerce management, and military security [1]. Synthetic aperture radar (SAR) is an attractive option for ship detection [2]–[6], [9]–[18], since: 1) it offers large swath widths to achieve extensive surveillance; 2) it is an active imaging technique with day-and-night surveillance ability; 3) it can image targets under clouds or haze. Note that SAR brings the abovementioned benefits for the task of ship detection, as well as for meeting some challenges, such as: 1) speckle noise [46] (caused by the interference on the echo from the surface elements of targets) and 2) complex background clutter [47], e.g., strong clutter (low contrast between sea clutter and weak targets) and heterogeneous clutter (caused by internal waves, breaking waves, sea currents, range/azimuth ambiguities, and so on). In recent years, many algorithms based on SAR images have been proposed to tackle these challenges and improve the performance of ship detection [2]–[6], [9]–[18], [25], [26].

Superpixels (SPs) represent perceptually coherent areas in the image. SPs can effectively capture the region-based features and reduce the loss of target detection performance caused by speckle noise in SAR images [7]–[9]. Several SP-based detectors derived from constant false alarm rate (CFAR) strategies [10]–[12] and derived kernels [13] have been developed to reduce false alarms in the detection result. But the detectors in [10]–[13] mainly focus on the first-order features of SPs and do not exploit the multiorder features therein.

Fisher vector (FV) is a recently introduced and powerful encoding strategy for SPs in SAR images. FVs contain the zero-order, first-order, and second-order features of SPs [16]–[20], [35]. In [17], the Euclidean distance is used to measure the local contrast of FVs (LCFVs), which lead to a higher detection rate in low signal-to-clutter ratio (SCR) scenarios. However, in heterogeneous clutter scenarios, the LCFV may produce some false alarms as it does not model the bright target parts in SAR images. It is also worth mentioning that the above FV-based ship detectors [17] (and also other ship detectors in [2]–[6], [10]–[15], and [18]) mainly focus on the local contrast of image features by using a sliding-window strategy, but ignore the cues from the global SAR images, which are important to make an accurate perception decision for humans.

According to human attention theory [21], the first step of visual attention is to obtain global cues based on "preattentive patterns." In this step, some image features are processed over

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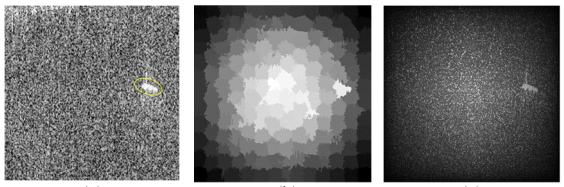
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(a) (b) (c)

Fig. 1. (a) Gaofen-3 SAR image, where a ship target is highlighted by using a yellow ellipse. (b) Global cue from (a) using the histogram feature in [22]. (c) Global cue from (a) using the directional feature in [23].

the entire image, providing important guidelines to help us overlook most of the nonsalient and noninformative background regions. In a second step, the "attentive patterns" are used to focus on a few salient regions in the image, based on local cues. Inspired by the "preattentive patterns" in human attention, the exploitation of global cues has been recently investigated in the context of optical image processing. In [22]–[24], global cues have been extracted based on the histogram and directional features to segment salient regions in man-made optical images. In Fig. 1, we show two global cues of a marine SAR image derived from histogram [22] and directional [23] features. As shown in Fig. 1(b) and (c), the global cues in [22] and [23] both provide degraded performance when discriminating target and clutter regions. This is because of the following reasons.

1) These two global cues are not robust to the speckle noise or the strong clutter that often contaminates marine SAR images but hardly exists in man-made optical images.
2) These two global cues mainly emphasize the regions close to the image centers that are often occupied by the targets of interest in man-made optical images [see [22, Fig. 5(a)]]. However, ship targets do not often locate at the centers of SAR images.

Therefore, the global cues of man-made optical images in the aforementioned works [22]–[24] are difficult to transfer into marine SAR images.

The research conducted on global cues in the context of ship target detection in SAR images is still quite limited. In [25] and [26], two global cues are developed and then combined with traditional local cues, which further provide the improved capability to discriminate between ship targets and the background clutter. However, the global cues in [25] and [26] ignore the density difference between ship target SPs and sea clutter SPs. For a given data point, the density reflects the number of data points close to it in feature space. In general, ship target SPs have much lower densities than sea clutter SPs, due to the sparse distribution of ship targets in the global SAR image [32]. The existing global cues in [25] and [26] may retain many background clutter areas with higher densities than target areas for the subsequent extraction of local cues, thus leading to an increase of false alarms in the final detection results.

Recently, density-based features have received extensive attention in target detection and classification problems in the context of biological analysis [27], hyperspectral image processing [28], [29], and video processing [30], due to their effectiveness and the possibility of fast implementation.

TABLE I

CATEGORIZATION OF RELATED WORKS ON SP-BASED SHIP TARGET DETECTION. LEGEND: ■INTENSITY FEATURE; ▲STATISTICAL FEATURE; ★DENSITY FEATURE

Paper	Local Cues	Global Cues
[11], 2019; [12], 2020; [14], 2016; [15], 2020; [16], 2020; [17], 2020; [18], 2020	▲	-
[10], 2016; [13], 2019	■	-
[25], 2016	■	■
[26], 2018	▲	▲
This paper (Improved performance of target enhancement and clutter suppression)	■+▲	▲+★

Note that ship targets sharing sparsity in spatiality are more like outlier data points (away from the "normal" clutter point set) in the feature space. However, commonly used density-based features [31] suppress outlier data points and cannot distinguish ship targets from the background clutter well.

In this article, we propose a new ship target detection method based on global and local cues (GLCs) derived from FVs (called GLCFV) that provides fewer false alarms than existing methods in the case of strong and heterogeneous clutter scenarios. In our newly developed GLCFV, the SPs in the SAR image are first obtained as basic processing units. Then, FVs are calculated as encoding vectors of SPs to capture their multiple order characteristics. Next, we introduce two new density features of FVs as global cues for ship detection. These new features are as follows.

1) *Reversed Density of FVs:* This feature is based on the principle that the density of ship target FVs is much lower than that of sea clutter FVs.
2) *Density-Based FV Distance:* This feature relies on the principle that ship target FVs exhibit large distances to sea clutter FVs with higher densities.

These two new global cues provide informative guidelines to suppress the background clutter areas and pick fairly small target areas in the global image. In GLCFV, we also develop the weighted LCFV as a local cue for ship detection, in which the original LCFV in [17] is improved by further highlighting the bright target areas with large intensities. The above two new global cues and an enhanced local cue are combined to generate the test statistic map of our newly developed GLCFV. Finally, thresholding is performed on the test statistic map to obtain the detection results.

As described in Table I, compared with existing (commonly used) SP-based ship detectors in [10]–[18], [25], and [26], the global density features of FVs are innovatively considered in the proposed method to suppress the global sea clutter areas and further reduce the false alarm rate. Besides, the existing local cue in [17] is modified by further enhancing the bright ship targets to reduce false detections. Unlike the traditional density features in [31], the densities of outliers are exploited in this work to highlight sparse ship targets. Our experiments with 1724 Gaofen-3 SAR image chips confirm that the proposed GLCFV ship detection approach provides better

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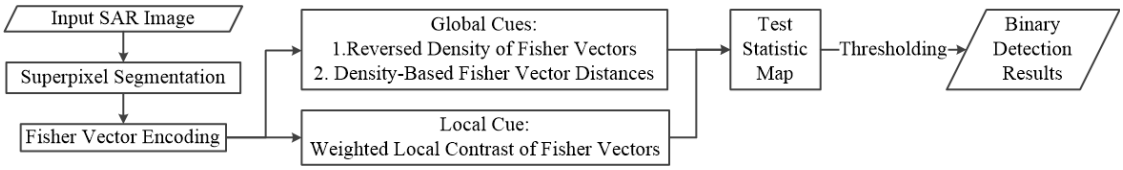


Fig. 2. Flowchart of our newly proposed GLCFV ship detection algorithm.

detection performance than other approaches, especially in scenarios with strong and heterogeneous sea clutter.

We also note that deep learning techniques have been recently investigated in the context of ship target detection [42]–[44]. In [42], a contextual region-based convolutional neural network (CNN) is proposed to improve the detection performance of ship targets with small sizes. Nieto-Hidalgo *et al.* [43] develop a two-stage CNN to achieve coarse-to-fine detection of ship targets. In [44], an anchor-free method, CenterNet, is proposed to detect ship targets in large-scale SAR images and reduce the false alarms caused by land areas. The abovementioned deep learning-based methods provide superior detection performance but require a large number of training data with labels. Besides, the interpretability of features captured by deep neural networks is still an open problem [45]. In this article, the proposed GLCFV approach considers the detection problem without training data, like [10]–[15], [17], [18], [25], and [26].

The remainder of this article is organized as follows. In Section II, we introduce the newly proposed GLCFV ship detector. Section III provides in-depth comparisons among the proposed GLCFV and other existing detectors via detailed experiments. Section IV concludes this article and provides some hints at plausible future research.

II. Proposed GLCFV Ship Detector

In this section, we propose a new ship detector to exploit global cues derived from FV density features, in addition to local cues derived from improved LCFVs. An overview of the proposed ship detection method is given in Fig. 2.

Preprocessing: SP Segmentation. First, SP segmentation is performed on the original SAR images to group spatially coherent pixels. SPs lead to a significant reduction of primitives in SAR images and increased computational efficiency. Besides, SPs reduce the effect of speckle noise in SAR images. Recently, several segmentation algorithms have been proposed to obtain SPs in SAR images [7], [9], [36], [49]. In [7], the simple linear iterative clustering (SLIC) algorithm is proposed according to a local k-means clustering strategy to achieve fast SP segmentation. Multiorder features [9] of SAR images are incorporated into SLIC to enhance the segmentation performance in low SCR cases. In [36], an edge detector that is robust to speckle noise in SAR images is adopted to improve the SP segmentation performance in SAR images. A finite mixture model-based SP (MISP) segmentation method is proposed in [49], generating regular SPs with smooth boundaries in SAR images. In the preprocessing stage of the proposed method, the SLIC algorithm in [7] is used to obtain SPs. We refer interested readers to [7] for a complete description of SLIC. Note that other state-of-the-art methods, such as [9], [36], and [49] for SP segmentation of SAR images, can also be used here to generate more compact SPs robust to speckle noise and to further improve the SP-based detection performance of ship targets. In our context, we select SLIC due to its simplicity [7].

Let M denote the number of SPs produced by SLIC in the input SAR image. The mth SP obtained by SLIC is represented by $\{\delta_{m,1}, \delta_{m,2}, \ldots, \delta_{m,L_m}\}$, where $\delta_{m,l}$ denotes the intensity of the lth pixel in the mth SP, L_m is the number of pixels therein, $l = 1.2,\ldots, L_m$, and $m = 1.2,\ldots, M$. Note that SLIC requires a predefined size of SPs in SAR images. In this article, the size of SPs is denoted by Q. An SP approximately covers Q^2 pixels with the setting of SP size Q. A discussion on the choice of Q will be reported in Section III.

A. Step 1: FV Encoding for SPs in SAR Images

In this section, the FV is introduced as a powerful encoding strategy to capture multiple order characteristics of SPs in SAR images. FVs are obtained by fitting a generative Gaussian mixture model (GMM) to the SPs to be encoded. Let $\boldsymbol{\lambda} = \{\omega_k, \mu_k, \sigma_k, k = 1, 2, \ldots, K\}$ represent the set of weights, mean values, and standard deviations in the GMM of the input SAR image, where K is the number of Gaussian components in the GMM. The set of parameters in $\boldsymbol{\lambda}$ can be estimated using the expectation-maximization algorithm [33], derived from a maximum likelihood estimation (MLE) strategy. The FV $\boldsymbol{\varphi}_m \in \mathbb{R}^{3K\times 1}$ of the mth SP is calculated as the gradient vector of the log-likelihood function with respect to the parameters in $\boldsymbol{\lambda}$ [19]

$$\boldsymbol{\varphi}_m = \left[\varphi_{m,\omega_1}, \ldots, \varphi_{m,\omega_K}, \varphi_{m,\mu_1}, \ldots, \varphi_{m,\mu_K}, \varphi_{m,\sigma_1}, \ldots, \varphi_{m,\sigma_K}\right]^{\mathrm{T}} \tag{1}$$

where

$$\varphi_{m,\omega_k} = (1/L_m)\sum_{l=1}^{L_m}\left(\phi_{m,l}^{(l)} - \beta_k\right)/\sqrt{\beta_k} \tag{2}$$

$$\varphi_{m,\mu_k} = (1/L_m)\sum_{l=1}^{L_m}\phi_{m,l}^{(k)}\left[\left(\delta_{m,l} - \mu_k\right)/\left(\sigma_k\sqrt{\beta_k}\right)\right] \tag{3}$$

$$\varphi_{m,\sigma_k} = (1/L_m)\sum_{l=1}^{L_m}\left[\phi_{m,l}^{(k)}/\sqrt{2\beta_k}\right]\left[\left(\delta_{m,l} - \mu_k\right)^2/\sigma_k^2 - 1\right] \tag{4}$$

and $m = 1.2,\ldots, M$. In (2)–(4), $\beta_k = \exp(\omega_k)/\sum_{\tilde{k}=1}^{K}\exp(\omega_{\tilde{k}})$ is the soft-max formulation of the weight parameter ω_k, and $\phi_{m,l}^{(k)} = \omega_k\mathcal{N}(\delta_{m,l};\mu_k,\sigma_k^2)/\sum_{\tilde{k}=1}^{K}\omega_{\tilde{k}}\mathcal{N}(\delta_{m,l};\mu_{\tilde{k}},\sigma_{\tilde{k}}^2)$ is the soft assignment of the pixel intensity $\delta_{m,l}$ to the kth Gaussian component denoted by $\mathcal{N}(\delta;\mu_k,\sigma_k^2)$. Then, signed square-rooting and self-l_2-normalization are conducted on each FV to enhance its performance [34]. From (2) to (4), we have

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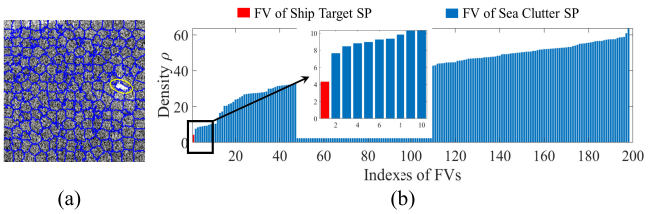


Fig. 3. (a) SP map of Fig. 1(a), and (b) associated density values of the FVs of SPs in (a) in ascending order, where the target FV has a lower density value than all the clutter FVs.

that the elements in the FV capture the zero-order, first-order, and second-order differences between the current SP and each Gaussian component in the GMM, which are beneficial to enhance the discrimination between ship targets and the background clutter [17].

B. Step 2: Global Cues Based on the Densities of FVs

In this section, we propose two new features based on the densities of FVs to hint at the global cues for the task of ship target detection. The density value of an FV reflects the aggregation level of other FVs close to it in terms of the Euclidean distance, i.e., the number of other FVs that are similar to it. Given the FV set $\{\boldsymbol{\varphi}_m, m = 1, 2, \ldots, M\}$ of all the SPs in the input SAR image, the density ρ_m of the mth FV is defined based on a Gaussian kernel function

$$\rho_m = \sum_{\tilde{m}\in\{1,2,\ldots,M\}/m} \exp\left[-\left(\frac{\mathcal{D}_{m,\tilde{m}}}{\mathcal{D}_c}\right)^2\right] \tag{5}$$

where

$$\mathcal{D}_{m,\tilde{m}} = \left\|\boldsymbol{\varphi}_m - \boldsymbol{\varphi}_{\tilde{m}}\right\|_2 \tag{6}$$

measures the Euclidean distance between the mth FV and the $\tilde{m}$th FV, $m \neq \tilde{m}$, and $\mathcal{D}_c > 0$ is the soft-cutoff distance. The density ρ_m is a larger value when the mth FV is surrounded by more FVs similar to itself, while it is a smaller value when the mth FV has larger Euclidean distances to the principal parts of all the FVs in the SAR image.

Global Cue 1: Reversed Density of FVs. Ship target SPs can be regarded as outliers in the global SAR image, due to the sparsity of ship targets in the spatial domain [32]. In other words, the number of target SPs is much smaller than that of sea clutter SPs in a SAR image. Accordingly, the density of target FVs is much lower than that of sea clutter FVs, as shown in Fig. 3. Based on the definition of the density in (5), we introduce the (normalized) reversed density features of FVs as follows:

$$f_m = 1 - \mathrm{Nom}(\rho_m) \tag{7}$$

where $\mathrm{Nom}(\cdot)$ denotes the normalized operation to let $f_m \in (0, 1)$, and $m = 1.2,\ldots, M$. Given a set of scalar values $\{a_m, \forall m\}$, $\mathrm{Nom}(\cdot)$ is defined as

$$\mathrm{Nom}(a) = \frac{a - \min\{a_m, \forall m\}}{\max\{a_m, \forall m\} - \min\{a_m, \forall m\}}. \tag{8}$$

As shown in (7), the densities of FVs are first normalized to [0, 1] and then reversed. In this way, target FVs with lower densities than sea clutter FVs are emphasized and larger values of f_m are assigned to them. On the contrary, sea clutter FVs are relatively suppressed in the feature map of $\{f_m, m = 1, 2, \ldots, M\}$.

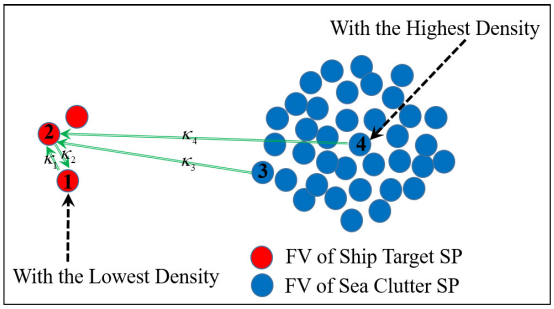


Fig. 4. 2-D plane to provide a sketch of κ in (10). The circle points represent the FVs of SPs in an image. Note that the relative locations of circle points do not represent those of SPs in an image. The Euclidean distance between two FVs of SPs in an image is represented by the distance between two circle points in this 2-D plane. The lengths of the green lines with arrows denote the values of κ belonging to the starting points.

Global Cue 2: Density-Based FV Distance. Due to the dissimilarity of multiple order information between ship target SPs and sea clutter SPs in the global SAR image, ship target FVs exhibit large distances to sea clutter FVs. Based on the above property of FV distances and the definition of density in (5), (normalized) density-based FV distance features are designed as follows:

$$g_m = 1 - \mathrm{Nom}(\kappa_m) \tag{9}$$

where

$$\kappa_m = \begin{cases} \max_{\tilde{m}\in\Lambda_m}\left\{\mathcal{D}_{m,\tilde{m}}\right\}, & \text{if } \Lambda_m \neq \emptyset \\ \min_{\tilde{m}\in\{1,2,\ldots,M\}/m}\left\{\mathcal{D}_{m,\tilde{m}}\right\}, & \text{if } \Lambda_m = \emptyset \end{cases} \tag{10}$$

and $\Lambda_m = \{\tilde{m}|\rho_{\tilde{m}} < \rho_m, \tilde{m} = 1, 2, \ldots, M\}$ contains the indexes of the FVs that exhibit lower densities than the mth FV.

Algorithm 1 GLCFV Ship Detector

Input : Test SAR image, the expected SP size Q, the number of Gaussian components in the GMM, K, and a soft-cutoff distance $\mathcal{D}_c$.
Preprocessing: Obtain the SP map of the input SAR image using the SLIC algorithm.
***Step* – 1**: Calculate FVs as the encoding vectors of SPs according to (1)-(4).
***Step* – 2**: Construct global cue maps based on the reversed density of FVs in (7) and the density-based FV distance in (9), respectively.
***Step* – 3**: Construct the local cue map via the weighted LCFV measure in (12).
***Step* – 4**: Combine global and local cues via (13), and then obtain binary detection results based on the threshold in (14).
Output: Binary ship detection results.

The density-based FV distance g_m in (9) is a reversed form of κ_m in (10). From (10), we can see that the value of κ_m is the maximum distance from the mth FV to other FVs with

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lower densities. If the mth FV has the lowest density, i.e., $\Lambda_m = \emptyset$, κ_m is the minimum Euclidean distance from the mth FV to other FVs. In Fig. 4, we provide an explicit sketch of κ in (10), where four representative FVs are highlighted. Obviously, the first FV has the lowest density value representing a typical target FV, while the highest density is occupied by the fourth FV belonging to clutter FVs. Having (10) in mind, the fourth FV has a large value of κ because of its large distance to the second FV with the lower density, i.e., $\kappa_4 = \max_{\tilde{m}\in\Lambda_4}\{\mathcal{D}_{4,\tilde{m}}\} = \mathcal{D}_{4,2}$. Similarly, the third clutter FV has a large value of κ, although it locates at the boundary of clutter FVs. On the contrary, the second target FV has a much smaller value of κ than the third and fourth clutter FVs due to the internal similarity of FVs among target SPs [16], i.e., $\kappa_2 = \max_{\tilde{m}\in\Lambda_2}\{\mathcal{D}_{2,\tilde{m}}\} = \mathcal{D}_{2,1}$. Due to the fact that $\Lambda_1 = \emptyset$, we have that $\kappa_1 = \min_{\tilde{m}\in\{1,2,\ldots,M\}/1}\{\mathcal{D}_{1,\tilde{m}}\} = \mathcal{D}_{1,2} = \mathcal{D}_{2,1} = \kappa_2$ is also a small value. Therefore, κ_m has a large value when the mth SP belongs to the clutter areas while it has a small value when a ship target occupies the mth SP. As a result, the density-based FV distances $\{g_m, m = 1, 2, \ldots, M\}$ in (9) highlight ship targets and weaken the global sea clutter by reversing the values of $\{\kappa_m, m = 1, 2, \ldots, M\}$ in (10).

The main difference between the FV density feature in (7) and the density-based FV distance feature in (9) is that the former considers the aggregation level of FVs while the latter exploits the dissimilarity between target FVs and clutter FVs with higher densities in Euclidean space. These two features enhance ship targets and suppress the global sea clutter using different viewpoints.

C. Step 3: Local Cue Based on Weighted LCFV

In this section, we introduce a weighted LCFV measure to extract a local cue for ship target detection. This local cue further suppresses the clutter areas that remain after the global cues, from a contextual perspective.

In general, local contrast is used to test if the current region is a saliency compared with its nearby regions. Recently, Wang *et al.* [17] have proposed an LCFV to distinguish between ship targets and their surrounding sea clutter in terms of multiple order statistics captured by FVs

$$\theta_m = \text{med}\left\{\left\|\boldsymbol{\varphi}_m - \boldsymbol{\varphi}_{m'}\right\|_2^2, \ \ \forall\, m' \in \mathcal{G}(m)\right\} \tag{11}$$

where med{.} represents the median value in a set of scalar values, $\mathcal{G}(m)$ is the set of nearby SPs of the mth one, and $m = 1.2,\ldots,M$. $\mathcal{G}(m)$ can be determined by using a sliding-window strategy, as shown in Fig. 1 of [12]. When the mth SP belongs to target areas, θ_m is a large value due to the difference of FVs between the target and its nearby clutter areas. On the other hand, θ_m tends to be a small value when the mth SP is located at the clutter areas, because of the similarity within clutter FVs. It has been demonstrated that the LCFV measure in (11) achieves good discrimination between targets and the background clutter in scenarios with low SCR.

Note that the LCFV in (11) does not consider *a priori* knowledge that ship target SPs are often the bright parts in SAR images, with higher average intensities than those of sea clutter SPs. This may cause some false alarms in the detection results. For example, in heterogeneous clutter scenarios, if there are perturbations in the FVs of some neighboring clutter SPs, then the most unusual clutter SPs therein may possess a large LCFV θ, similar to that of a true ship target, although these clutter SPs indeed may share lower intensities than true ship target SPs. In the following, we modify the LCFV in (11) by embedding the intensity contrast (as a weight to the Euclidean distance) to enhance its robustness to the heterogeneous clutter. The weighted LCFV is defined as

$$\tilde{\theta}_m = \text{med}\left\{\left\|\left(\varsigma_m/\varsigma_{m'}\right)\times\left(\boldsymbol{\varphi}_m - \boldsymbol{\varphi}_{m'}\right)\right\|_2^2, \ \ \forall\, m' \in \mathcal{G}(m)\right\} \tag{12}$$

where ς_m denotes the average intensity of the mth SP, and $m = 1, 2, \ldots, M$. When the mth target SP has a larger average intensity than its surrounding clutter SPs, the weights $\{\varsigma_m/\varsigma_{m'}, m' \in \mathcal{G}(m)\}$ are large values, i.e., $\varsigma_m/\varsigma_{m'} \gg 1$, for $\forall m' \in \mathcal{G}(m)$. On the contrary, we have $\varsigma_m/\varsigma_{m'} \approx 1$ when both the mth and the m'th SPs belong to sea clutter areas. This is helpful to enhance the performance of the original LCFV in (11) by highlighting the bright parts in the SAR images.

D. Step 4: Generation of Test Statistic Map by Fusing the GLCs

In this section, the global cues [reversed density of FVs in (7) and density-based FV distance in (9)] and the local cue [from the weighted LCFV in (12)] are combined to generate a final test statistic map

$$T_m = \underbrace{f_m \times g_m}_{\text{Global Cues}} \times \underbrace{\tilde{\theta}_m}_{\text{Local Cue}} \tag{13}$$

where $m = 1, 2, \ldots, M$. In (13), f_m emphasizes that ship targets exhibit a lower density of multiple order information captured by FVs than clutter areas. Besides, g_m extracts the saliencies of ship targets based on the dissimilarity between target FVs and clutter FVs and the difference of densities between them in the global SAR image. Both f_m and g_m enhance ship targets and suppress clutter areas throughout the input SAR image. The weighted LCFV $\tilde{\theta}_m$ exploits the local differences between target FVs and clutter FVs using *a priori* knowledge of intensity differences between target SPs and clutter SPs. In (13), the multiplication fusion is used due to the fact that it does not require parameter weighting.

Finally, ship targets are segmented from the test statistic map $\{T_m, m = 1.2,\ldots,M\}$ by using an adaptive threshold derived from the two-parameter method [37]

$$\tau = \tilde{\mu} + p\tilde{\sigma} \tag{14}$$

where $\tilde{\mu}$ and $\tilde{\sigma}$ are the mean value and the standard deviation of $\{T_m, m = 1.2,\ldots, M\}$, respectively, and p is an adjustment parameter often ranging from 3 to 5 [37]. If $T_m > \tau$, then the mth SP is occupied by ship targets. Otherwise, the mth SP belongs to the background clutter.

The implementation of the proposed GLCFV detection approach is summarized in Algorithm 1.

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TABLE II
CHARACTERISTICS OF THE SAR IMAGES USED IN OUR EXPERIMENTS

Satellite	Gaofen-3	
Polarization	HH	
Frequency Band	C	
Center Frequency	5.4GHz	
Incidence Angle	20°-50°	
Resolution	3m×3m	1m×1m
Imaging Mode	UFS	SL
Size	256×256	3000×3000
Number of Images	1723	1
Number of Ships	1935	20

III. EXPERIMENTAL RESULTS

This section validates the performance of our newly proposed GLCFV detector using metrics such as the target-to-clutter contrast and the receiver operating characteristic (ROC) curve, as compared to other state-of-the-art ship detectors, i.e., SP-based local contrast measure (SLCM) [13], LCFV [17], saliency and context information contrast (SCIC) [25], SP-based statistical dissimilarity (SSD) [26], curvature-based saliency detection (CBSD) [15], and truncated SP-CFAR (TSP-CFAR) [12], using SAR images from Gaofen-3 satellites [38], [39]. Here, the SAR images are acquired with different resolutions in high/low SCR and homogenous/heterogeneous clutter scenarios. Table II shows the corresponding parameters of the SAR images used in this section. The wind speed on the ocean is a key parameter affecting the state of the sea clutter [40], [41]. Although the exact wind speeds are not available in the open SAR image data set [38], [39], the effect of wind speed on the proposed ship detection method is examined by using SAR images with different sea states, as shown in Section III-A. The ground truth of ship targets in the considered SAR images is obtained from the annotations in [38] and [39].

Hyper-Parameter Settings: The GMM with a large value of K can fit SAR images well, but increases the computational burden of parameter pre-estimation for the set $\boldsymbol{\lambda} \triangleq \{\omega_k, \mu_k, \sigma_k, k = 1, 2, \ldots, K\}$. But if we use the GMM with a smaller K, e.g., $K = 2$, we cannot process SAR images with complex background clutter. In this paper, we set $K = 7$ as suggested in [16] and [17]. The soft-cutoff distance $\mathcal{D}_c$ is determined by the maximum value of all the FV distances $\{\mathcal{D}_{m,\tilde{m}}, \forall m, \tilde{m}\}$ in a SAR image, i.e.,

$$\mathcal{D}_c = \xi \times \max\left\{\mathcal{D}_{m,\tilde{m}}, \forall m, \tilde{m}\right\} \tag{15}$$

where $1 > \xi > 0$ is a constant. In this article, $\xi = 0.3$ is selected. A fixed percentage (e.g., 25%–100%) of the average number of pixels in the area occupied by a ship target is set as the expected SP size Q. In Section III-B, we evaluate the robustness of the proposed GLCFV detector to the constant ξ in (15) and the SP size Q, respectively.

A. Target-to-Clutter Contrast in Test Statistic Map

In this section, we illustrate the target enhancement and clutter suppression performance of the proposed GLCFV detector (as compared to that of other competitive detectors) in terms of visualization and quantitative contrast factors (CFs) [17]

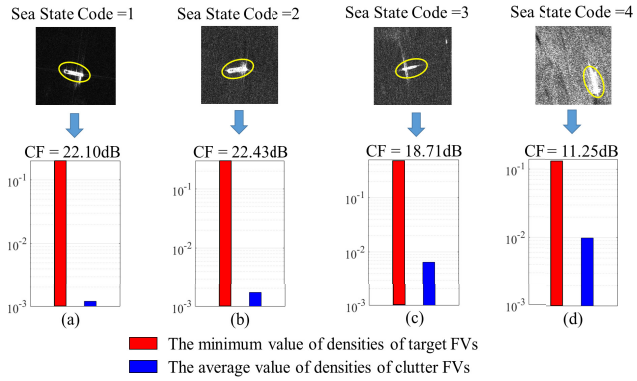


Fig. 5. Contrast between the minimum value of densities of target FVs and the average value of densities of clutter FVs. The sea state codes of the SAR images [39] in (a)–(d) are 1–4.

$$\mathrm{CF} \triangleq 10\log_{10}\frac{\min_{\bar{m}}\left\{T'_{\bar{m}}, \bar{m} = 1, 2, \ldots, N_{\mathrm{Truetar}}\right\}}{\frac{1}{|\mathrm{Clu}|}\sum_{m\in\mathrm{Clu}} T_m} \tag{16}$$

where N_{Truetar} denotes the number of ground truth ship targets in the SAR image, "Clu" and |Clu| represent the clutter region in the SAR image and the number of clutter SPs therein, respectively, and $T'_{\bar{m}}$ is the maximum value of the test statistics in the $\bar{m}$-th target area. In essence, CF measures the contrast between the weakest target and the global clutter level in the test statistic map. A large value of CF benefits the detection performance of weak ship targets.

A larger sea state code means a stronger sea clutter level. In Fig. 5, we show the contrast between the densities of target FVs and clutter FVs with different sea state codes. As shown in Fig. 5, the minimum value of the densities of target SPs is still much larger than the average value of densities of clutter SPs with the increase of sea state codes. This demonstrates the robustness of the proposed density feature of FVs to different sea state codes. We refer readers to [48] for a more detailed introduction of sea state codes.

Fig. 6 shows a SAR image chip with heterogeneous sea clutter, the corresponding SP map, and the test statistic maps obtained by the proposed GLCFV and other competitive methods. Each statistic map in Figs. 6–9 is normalized into the range [0, 1], using the Nom(.) operation in (8). From Fig. 6(c) and (d), we can see that SLCM [13] and LCFV [17] suffer from strong clutter areas, which may lead to false alarms in the final detection results. This is because SLCM [13] and LCFV [17] exploit only the local contrast cues without considering global cues. SSD [26] and SCIC [25] further embed the global contrast information into ship detectors and provide better performance in terms of target enhancement and background clutter suppression, as illustrated in Fig. 6(g) and (j). In Fig. 6(k) and (l), we show the global cues from the reversed density of FVs and the density-based FV distances, respectively. It can be observed from Fig. 6(k) that the reversed density of FVs in (7) can reveal the sparse target areas, but also presents some clutter areas. In Fig. 6(l), the density-based FV distance in (9) provides a clearer background than the reversed density of FVs in (7), since the contrast between target and clutter SPs is increased via the density-based FV distance.

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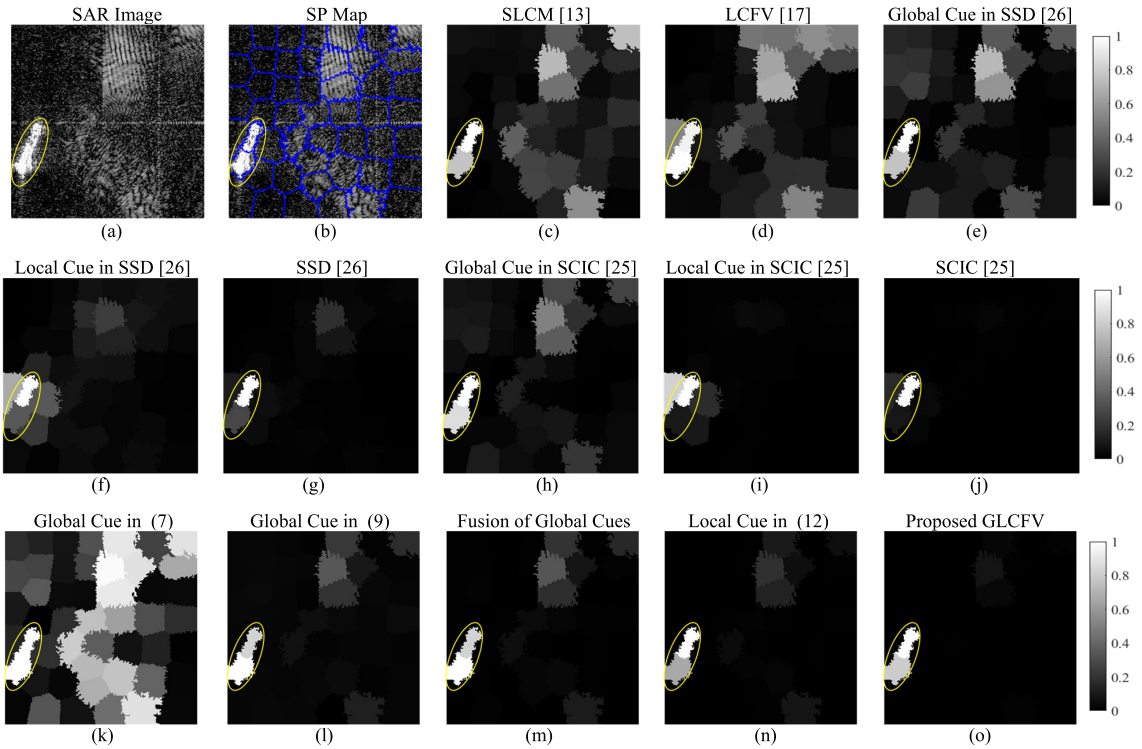


Fig. 6. (a) Marine SAR image with heterogeneous sea clutter. (b) Superpixel map with $Q = 32$. (c) Test statistic map of SLCM [13]. (d) Test statistic map of LCFV [17]. (e) Global cue in SSD [26]. (f) Local cue in SSD [26]. (g) Test statistic map of SSD [26]. (h) Global cue in SCIC [25]. (i) Local cue in SCIC [25]. (j) Test statistic map of SCIC [25]. (k) Global cue 1: reversed density of FVs in (7). (l) Global cue 2: density-based FV distance in (9). (m) Multiplicative fusion of global cues 1 and 2, see (13). (n) Local cue: weighted LCFV in (12). (o) Test statistic map of the proposed GLCFV.

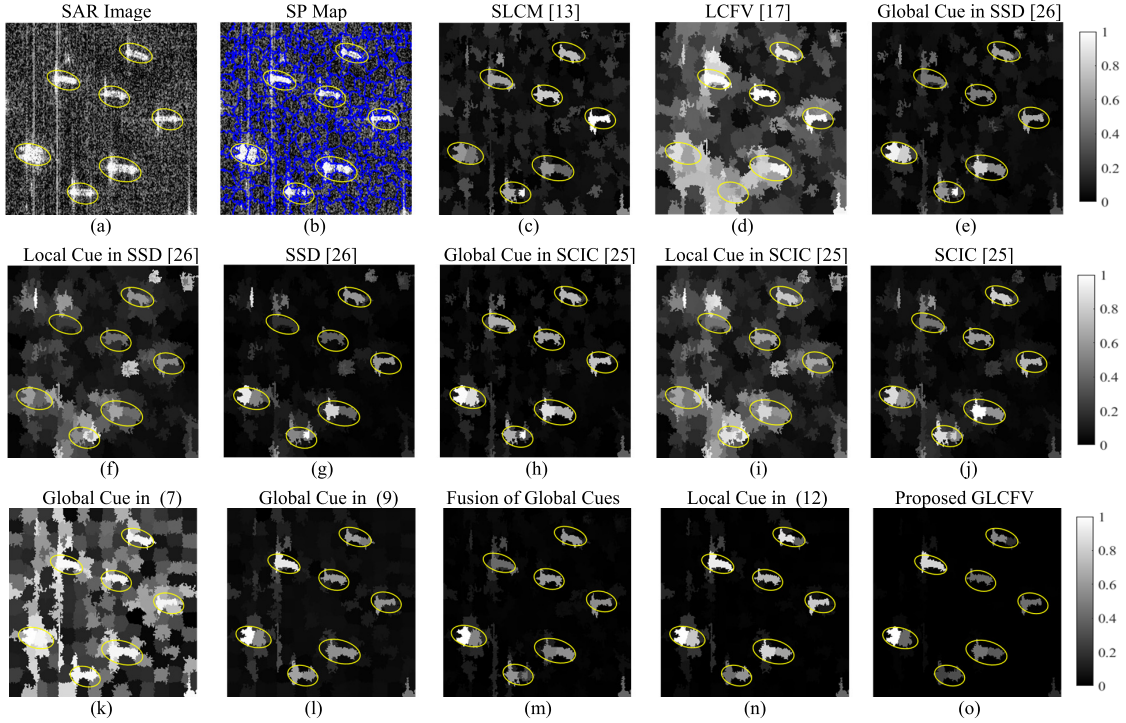


Fig. 7. (a) Marine SAR image with multiple ship targets. (b) Superpixel map with $Q = 18$. (c) Test statistic map of SLCM [13]. (d) Test statistic map of LCFV [17]. (e) Global cue in SSD [26]. (f) Local cue in SSD [26]. (g) Test statistic map of SSD [26]. (h) Global cue in SCIC [25]. (i) Local cue in SCIC [25]. (j) Test statistic map of SCIC [25]. (k) Global cue 1: reversed density of FVs in (7). (l) Global cue 2: density-based FV distance in (9). (m) Multiplicative fusion of global cues 1 and 2, see (13). (n) Local cue: weighted LCFV in (12). (o) Test statistic map of the proposed GLCFV.

In Fig. 6(m), we show the multiplicative fusion result of the two proposed global cues in (7) and (9), where a reduced sea clutter level can be observed when compared with the global cues in SSD [26] and SCIC [25], as shown in Fig. 6(e) and (h). Fig. 6(n) shows the weighted LCFV measure in (12), which provides improved image contrast compared with its original

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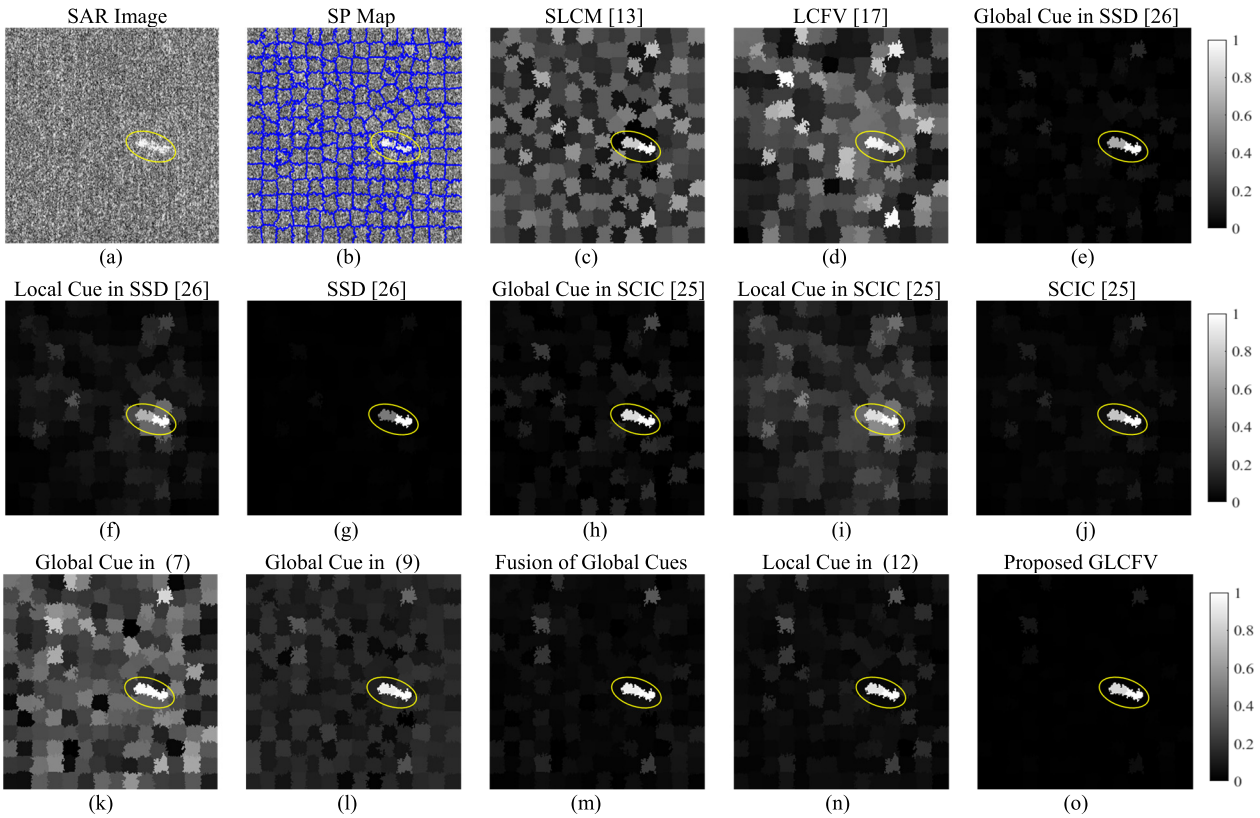


Fig. 8. (a) Marine SAR image with the small and faint ship target. (b) Superpixel map with $Q = 18$. (c) Test statistic map of SLCM [13]. (d) Test statistic map of LCFV [17]. (e) Global cue in SSD [26]. (f) Local cue in SSD [26]. (g) Test statistic map of SSD [26]. (h) Global cue in SCIC [25]. (i) Local cue in SCIC [25]. (j) Test statistic map of SCIC [25]. (k) Global cue 1: reversed density of FVs in (7). (l) Global cue 2: density-based FV distance in (9). (m) Multiplicative fusion of global cues 1 and 2, see (13). (n) Local cue: weighted LCFV in (12). (o) Test statistic map of the proposed GLCFV.

version, i.e., LCFV [17] in Fig. 6(d). This is mainly due to the newly introduced intensity-based weights, which are very helpful to reduce the false alarms in the (final) binary detection results. In Fig. 6(o), the proposed GLCFV ship detection method significantly highlights the ship target and reduces the heterogeneous clutter level.

In Fig. 7, we illustrate the contrast between target and clutter areas produced by GLCFV and the existing ship detectors when ship targets are densely distributed in the SAR image. As indicated in Fig. 7, both SSD [26] and SCIC [25] (with additional global contrast information) still provide better target-to-clutter contrast when compared to SLCM [13] and LCFV [17]. The map of the reversed density of FVs highlights all the ship targets, even in a complex multi-target scenario, but also highlights some clutter areas, like the scattering tails of ship targets with sparsity in spatiality. The maps of density-based FV distance and weighted LCFV emphasize better the target areas than the reversed density of FVs but still suffer from a few clutter areas that are not well suppressed. The fusion of the reversed density of FVs and the density-based FV distance still provides better target-to-clutter contrast than the global cues used in SCIC [25] and SSD [26]. In addition, the proposed weighted LCFV suppresses the sea clutter better than the LCFV [17]. As shown in Fig. 7(o), by combining the proposed GLCs from FVs, GLCFV provides clearer background than other competitive methods in the considered multi-target scenario.

Figs. 8 and 9, respectively, illustrate the target-to-clutter contrast in scenarios with strong sea clutter and interference of ground clutter. In Fig. 8, we can see that both the target and (a few) clutter areas are enhanced by SLCM [13] and LCFV [17]. SSD [26] and SCIC [25] produce clearer background than SLCM [13] and LCFV [17]. Similar to SLCM [13] and LCFV [17], the reversed densities of FVs highlight both the target and a few strong clutter SPs. The density-based FV distance and the weighed LCFV provide better performance in the task of reducing clutter than SLCM [13], LCFV [17], and the reversed density of FVs. The GLCFV successfully maintains the target areas and suppresses the strong sea clutter due to the combination of global density features and the weighted LCFV feature. From Fig. 9, it can be observed that ground clutter areas at the lower leftmost corner of the SAR image are retained in all the test statistic maps, due to their large intensities (which are very similar to those of the ship targets). Our experiments in Fig. 9 reveal that the proposed GLCFV method still provides better or comparable performance in the tasks of target enhancement and clutter reduction than the other tested methods.

Table III summarizes the CFs associated with the results in Figs. 6–9. As shown in Table III, the combination of two global cues in our GLCFV generates better (or comparable) CFs than other existing global cues. Specifically, the fusion of the two global cues outperforms the results obtained by each of them individually. The local cue in GLCFV, i.e., the

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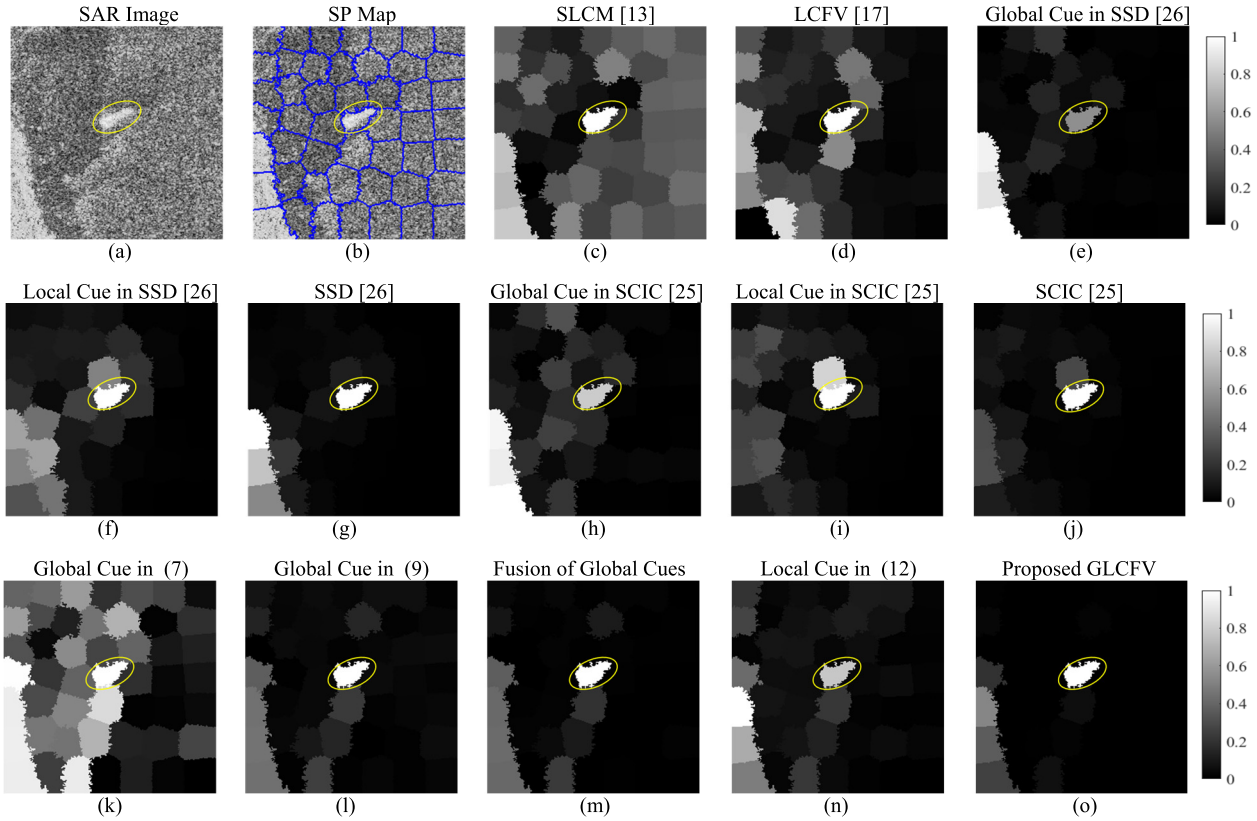


Fig. 9. (a) Marine SAR image contaminated by the ground clutter. (b) Superpixel map with $Q = 32$. (c) Test statistic map of SLCM [13]. (d) Test statistic map of LCFV [17]. (e) Global cue in SSD [26]. (f) Local cue in SSD [26]. (g) Test statistic map of SSD [26]. (h) Global cue in SCIC [25]. (i) Local cue in SCIC [25]. (j) Test statistic map of SCIC [25]. (k) Global cue 1: reversed density of FVs in (7). (l) Global cue 2: density-based FV distance in (9). (m) Multiplicative fusion of global cues 1 and 2, see (13). (n) Local cue: weighted LCFV in (12). (o) Test statistic map of the proposed GLCFV.

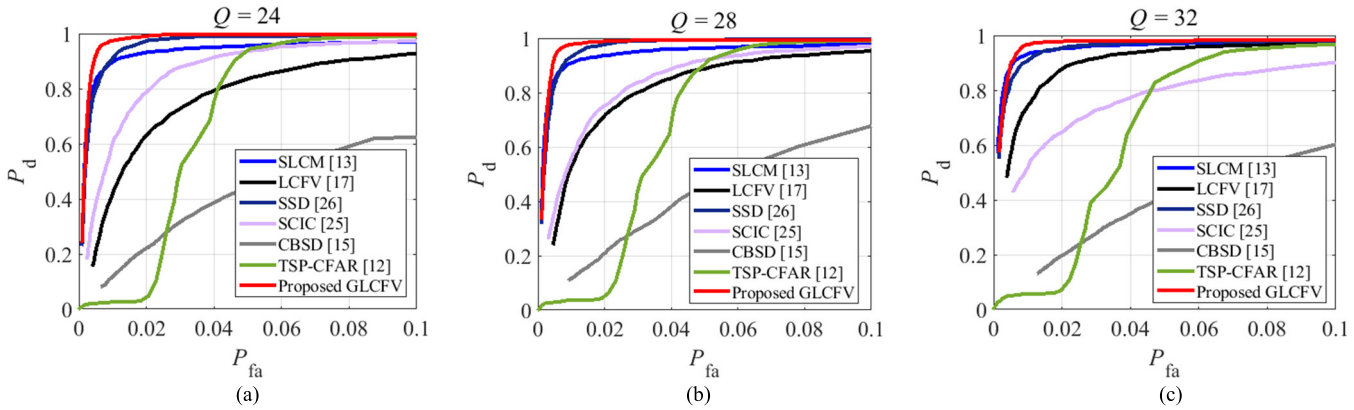


Fig. 10. ROC curves based on 1723 SAR images [38] with different SP sizes (Q) in SP generation. (a) $Q = 24$. (b) $Q = 28$. (c) $Q = 32$.

weighted LCFV, also exhibits better (or comparable) CFs than other existing local cues. Furthermore, higher CFs are obtained by the proposed GLCFV detection method. This confirms that GLCFV exhibits better clutter suppression characteristics than the other tested methods. This is very helpful to distinguish between ship targets and complex background clutter. It should be noted that the performance of CBSD [15] and TSP-CFAR [12] is not reported in Figs. 6–9 and Table III. This is due to the following reasons: 1) the SP map of CBSD [15] is different from that of other competitive methods since, in the CBSD [15], a nonlinear anisotropic diffusion filter is performed on the SAR images before the SP generation step and 2) the test statistic map of TSP-CFAR [12] cannot reflect the global target-to-clutter contrast since, in the TSP-CFAR [12], adaptive decision thresholds for different SPs are calculated instead of a global threshold.

B. SP-Based Detection Performance

In this section, we report the detection performance of TSP-CFAR [12], SLCM [13], CBSD [15], LCFV [17], SCIC [25], and SSD [26] using ROC curves. The probability of detection

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TABLE III
CFs (IN dB) ASSOCIATED WITH THE RESULTS IN FIGS. 6–9

	CFs			
	Fig. 6	Fig. 7	Fig. 8	Fig. 9
SLCM [13]	8.53	7.78	5.80	5.17
LCFV [17]	7.82	3.16	11.85	7.23
Global Cue in SSD [26]	9.27	7.44	**16.73**	7.85
Local Cue in SSD [26]	11.51	0.82	11.63	8.92
SSD [26]	17.30	4.31	23.23	11.38
Global Cue in SCIC [25]	11.47	10.48	13.85	7.99
Local Cue in SCIC [25]	15.42	2.65	8.13	**10.17**
SCIC [25]	22.05	7.34	19.39	13.50
Global 1: Reversed Density of FVs	4.63	3.75	5.02	4.71
Global 2: Density-Based FV Distance	12.99	9.19	8.50	11.07
Fusion of Global Cues in GLCFV	**15.13**	**11.74**	13.70	**12.33**
Local Cue in GLCFV	**17.67**	**12.35**	**12.26**	8.58
Proposed GLCFV	**24.50**	**18.83**	**24.03**	**14.57**

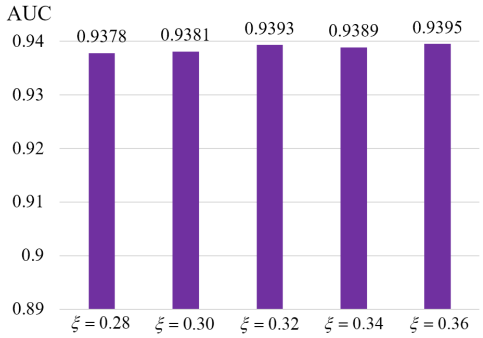


Fig. 11. AUC values of the proposed GLCFV detector based on 1723 SAR images [38] with different values of ξ in (15).

and the probability of false alarm therein are calculated as $P_{\mathrm{d}} = N_{\mathrm{Dtar}}/N_{\mathrm{Truetar}}$ and $P_{\mathrm{fa}} = N_{\mathrm{Dfal}}/N_{\mathrm{clu}}$, respectively, where N_{Dtar} and N_{Dfal} represent the number of detected ship targets and false alarm SPs, respectively, and N_{clu} is the number of clutter SPs in the original SAR images.

In Fig. 10, we first show the ROC curves corresponding to the 1723 SAR images contained in the data set described in [38], where some examples in this data set have been presented in Figs. 6(a)–9(a) for illustrative purposes. From Fig. 10, we can see that SSD [26] (combining GLCs) maintains a better detection performance (with different SP sizes) than TSP-CFAR [12], SLCM [13], and LCFV [17], which are derived from the local contrast cues. SCIC [25] also exploits GLCs from SAR images, but provides lower detection capability than the SSD [26]. CBSD [15] suffers from low detection performance since the nonlinear anisotropic diffusion filter in CBSD reduces the intensities of small ship targets. It is remarkable that, in comparison with other competitive ship detectors, the proposed GLCFV detector improves the probability of detection P_{d} (with a fixed probability of false alarm P_{fa}) as compared with other methods. Fig. 10 also confirms the robustness of the proposed GLCFV approach to mild changes of SP sizes.

Fig. 11 shows the area under the ROC curve (AUC) values of the proposed GLCFV detector, where different values of ξ are used to determine the soft-cutoff distance $\mathcal{D}_c$. We can see that GLCFV produces consistently high AUC values with different soft-cutoff distances. This is because GLCFV mainly relies on the relative magnitude of densities instead of the soft-cutoff distance.

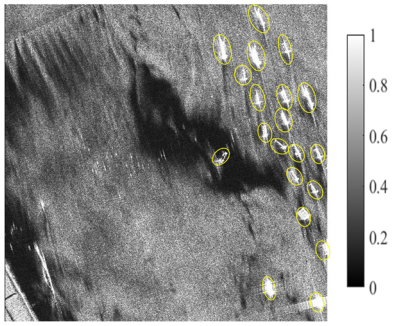


Fig. 12. Large SAR image (3000 × 3000) [39] with 20 ship targets and heterogeneous sea clutter. The sea state code is 4.

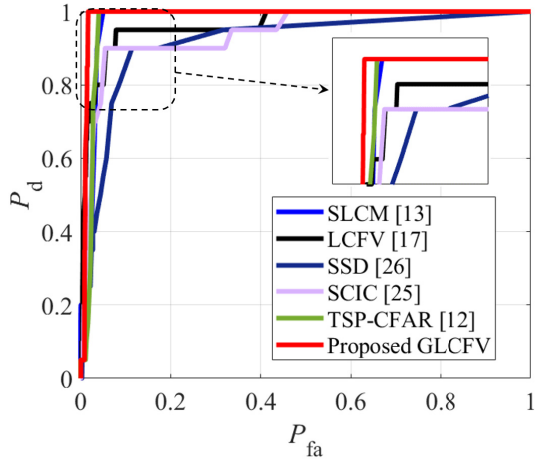


Fig. 13. ROC curves for the image in Fig. 12 with $Q = 70$.

TABLE IV
AUC VALUES FOR THE IMAGE IN FIG. 12 VERSUS SP SIZE

	AUC		
	$Q = 70$	$Q = 74$	$Q = 78$
SLCM [13]	0.9771	0.9813	0.9842
LCFV [17]	0.9633	0.9592	0.9665
SSD [26]	0.9167	0.9394	0.9389
SCIC [25]	0.9407	0.9498	0.9432
TSP-CFAR [12]	0.9757	0.9750	0.9717
Proposed GLCFV	**0.9891**	**0.9870**	**0.9858**

In Fig. 12, we show a large SAR image with 20 ship targets. The sea state code of Fig. 12 is 4, which is the largest one in the SAR data set [39], meaning complex background clutter in the considered SAR images. Fig. 13 shows the ROC curves of the proposed GLCFV and other existing methods for the case study in Fig. 12. In Fig. 13, the proposed GLCFV detector is shown to be superior to other existing methods. This is mainly due to the newly exploited global density features and the improved LCFV measure. Because CBSD shows degraded performance with complex backgrounds in SAR images, its ROC is not included in Fig. 13. In Table IV, the proposed GLCFV method is shown to provide higher AUCs than other methods (for different SP sizes) when applied to the large SAR image in Fig. 12. This demonstrates the effectiveness of GLCFV in the task of processing very large SAR images.

It is worth mentioning that the shape and size of ship targets may not be well-preserved in SP-based detection results generated by the proposed and other existing detection

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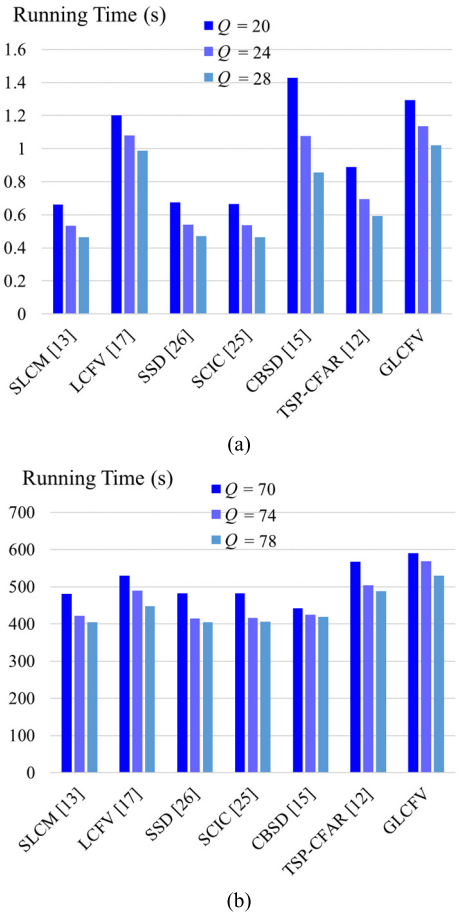


Fig. 14. (a) Average running times based on 1723 SAR images with SP sizes $Q = 20$, 24, and 28, where each image contains 256×256 pixels. (b) Running times based on the SAR image in Fig. 12 containing 3000×3000 pixels with SP sizes $Q = 70$, 74, and 78.

methods. For example, a ship target may be separated into two or three SPs, as shown in Figs. 6 and 8. To address the above issue, SP merging algorithms [8] can be used to guarantee the integrity of ship targets. We refer readers to [8] for a more detailed description of SP merging, since a thorough discussion on this topic is beyond the scope of this work.

C. Running Time

In Fig. 14, we compare the running times of the proposed GLCFV method with that of other competitive methods. The software platform is MATLAB 2019b in a 64-bit Windows system with an Intel(R) Xeon(R) platinum 8280 CPU and 1-TB of RAM. From Fig. 14, we can see that the running times of all the SP-based detection methods are reduced with the increase of SP size. The running times of TSP-CFAR [12], SLCM [13], SCIC [25], and SSD [26] are smaller than those of GLCFV and LCFV [17], mainly because the calculation of FVs in the last two methods increases the computational burden. The running time of GLCFV is higher than that of LCFV, mainly due to the additional exploitation of global cues. Compared with other methods, we can see that GLCFV provides slightly larger but comparable running times.

IV. Conclusion

In this article, we have introduced a new GLCFV detector for ship detection in marine SAR images. FVs are used to encode the multiple order information of SPs. Our method incorporates two simple but powerful global cues, derived from the reversed density of FVs and the density-based FV distances, respectively, which greatly enhance the discrimination between ship targets and the background clutter. The reversed density of FVs reflects the aggregation level of FVs, where the density of ship target FVs is far lower than that of clutter FVs, due to the sparsity of ship targets in the global SAR image. The density-based FV distance measures the dissimilarity between target FVs and clutter FVs (with higher density values than the former). The local cue in GLCFV is an improved version of LCFV, where the intensity contrast between ship targets and background clutter is embedded into LCFV, for the first time in the literature, to reduce false alarms in the final detection results. Our experiments with a large database of Gaofen-3 SAR images demonstrate that the proposed GLCFV detector dramatically improves the detection performance in comparison with existing detectors, especially in scenarios with heterogeneous and strong background clutter. The exploitation of density-based features in ship target classification problems will be the topic of our future research. Another topic deserving future research is to explore the possible weighting of the three considered cues (this would involve an expensive optimization problem that may require parallelization and/or hardware implementation).

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